THE MASTERMIND COACHING CLASSES

CBSE / RBSE · Nursery – Class XII · Road No. 5, VKIA, Jaipur · 9887258679

ANSWER KEY & MARKING SCHEME — MATHS · CLASS 9 · SET 03

Max. Marks: 80 · The Mastermind Coaching Classes, Jaipur · Alternate correct methods to be given full credit

SECTION A — Objective Type — 20 questions · 20 marks

Q1–Q18 are multiple-choice questions (1 mark each). Q19–Q20 are Assertion (A) and Reason (R) questions: select the correct option from (a), (b), (c), (d) given below each.

1. The point on the y-axis, 5 units below the x-axis, is:

(a) (0, 5) (b) (5, 0) (c) (−5, 0) (d) (0, −5)

Ans: (d) On the y-axis abscissa = 0; below ⇒ ordinate −5.

2. The point (−3, −8) lies in the:

(a) second quadrant (b) first quadrant

(c) third quadrant (d) fourth quadrant

Ans: (c) Both coordinates are negative.

3. The distance of the point (−5, 12) from the origin is:

(a) 17 units (b) 7 units (c) √119 units (d) 13 units

Ans: (d) √(25 + 144) = 13.

4. The degree of the polynomial $2x^5 - x^3 + 9$ is:

(a) 3 (b) 5 (c) 2 (d) 9

Ans: (b) Highest power of x is 5.

5. The length and breadth of a rectangle are (x + 4) and (x − 1) respectively. Its area is:

(a) $x^2 + 3x - 4$ (b) $x^2 + 5x - 4$ (c) $2x + 6$ (d) $x^2 - 4$

Ans: (a) $(x + 4)(x - 1) = x^2 + 3x - 4$.

6. A linear polynomial p(x) satisfies p(0) = 3 and p(2) = 7. Then p(x) is:

(a) 2x + 3 (b) x + 5 (c) 2x − 3 (d) 3x + 2

Ans: (a) p(0) = 3 ⇒ constant 3; p(2) = 7 ⇒ 2a + 3 = 7 ⇒ a = 2.

7. Which of the following is an irrational number?

(a) $\frac{22}{7}$ (b) √1.69

(c) $\frac{\sqrt{2}}{2}$ (d) 0.333… (3 repeating)

Ans: (c) √2/2 = 1/√2 is irrational; √1.69 = 1.3 and the others are rational.

8. The value of |−3.5| + |2.5| is:

(a) 6 (b) −1 (c) 1 (d) −6

Ans: (a) 3.5 + 2.5 = 6.

9. Which of the following rational numbers lies between $\frac{3}{5}$ and $\frac{4}{5}$?

(a) $\frac{7}{10}$ (b) $\frac{9}{10}$ (c) $\frac{4}{5}$ (d) $\frac{1}{2}$

Ans: (a) 3/5 = 0.6 < 0.7 < 0.8 = 4/5.

10. The factorisation of $x^2 + 10x + 25$ is:

(a) $(x + 5)^2$ (b) $(x + 25)(x + 1)$
(c) $(x + 5)(x - 5)$ (d) $(x - 5)^2$

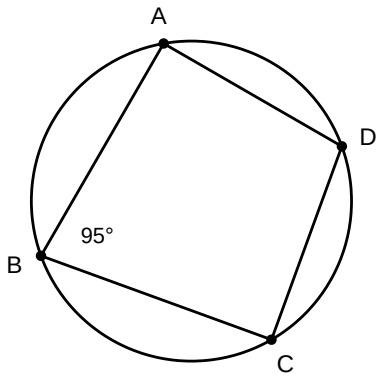
Ans: (a) $x^2 + 2(x)(5) + 5^2$.

11. If $x + y = 8$ and $xy = 12$, then $x^2 + y^2$ is equal to:

(a) 20 (b) 52 (c) 40 (d) 64

Ans: (c) $x^2 + y^2 = (x + y)^2 - 2xy = 64 - 24 = 40$.

12. In the figure ABCD is a cyclic quadrilateral with $\angle ABC = 95°$. Then $\angle ADC$ is:



(a) 75° (b) 85° (c) 95° (d) 170°

Ans: (b) Opposite angles are supplementary: $180° - 95° = 85°$.

13. A chord of a circle of radius 13 cm is at a distance of 12 cm from the centre. The length of the chord is:

(a) 12 cm (b) 5 cm (c) 25 cm (d) 10 cm

Ans: (d) Half chord = $\sqrt{(169 - 144)} = 5 \Rightarrow$ chord = 10 cm.

14. The area of a triangle with sides 9 cm, 40 cm and 41 cm is:

(a) 360 cm^2 (b) 90 cm^2 (c) 369 cm^2 (d) 180 cm^2

Ans: (d) $s = 45$; area = $\sqrt{(45 \times 36 \times 5 \times 4)} = \sqrt{32400} = 180$.

15. The sides of a triangular field are in the ratio 3 : 5 : 7 and its perimeter is 150 m. The shortest side is:

(a) 50 m (b) 10 m (c) 30 m (d) 45 m

Ans: (c) One part = $150 \div 15 = 10$ m; $3 \times 10 = 30$ m.

16. A bag has 3 red, 5 blue and 2 green balls. A ball is drawn at random. The probability that it is not blue is:

(a) $\frac{1}{5}$ (b) $\frac{1}{10}$ (c) $\frac{3}{10}$ (d) $\frac{1}{2}$

Ans: (d) Not blue = 3 + 2 = 5 out of 10.

17. A number is chosen at random from the numbers 1 to 20. The probability that it is a multiple of 5 is:

(a) $\frac{1}{5}$ (b) $\frac{3}{20}$ (c) $\frac{1}{20}$ (d) $\frac{1}{4}$

Ans: (a) 5, 10, 15, 20 $\Rightarrow$ 4/20.

18. If $u_n = (-1)^n(3n - 2)$, then u_4 is:

(a) 14 (b) −14 (c) −10 (d) 10

Ans: (d) $(-1)^4 \times 10 = 10$.

19. **Assertion (A):** The number 0.1010010001… (with increasing number of zeros) is an irrational number.
Reason (R): A non-terminating and non-repeating decimal is an irrational number.

Ans: (a) The decimal never ends and has no repeating block, so it is irrational; R states the reason.

20. **Assertion (A):** The polynomial $3x - 5$ has exactly one zero.
Reason (R): A linear polynomial $ax + b$ ($a \neq 0$) has exactly one zero, which is $-b/a$.

Ans: (a) Zero = 5/3; R is the general result.

SECTION B — Very Short Answer (2 marks each) — 5 questions · 10 marks

Internal choice is provided in one question.

21. A cab charges ₹60 for the first 2 km and ₹12 for every additional km. Find the fare for a journey of 15 km.

Additional distance = 13 km (½); extra charge = 13 × 12 = ₹156 (1); fare = 60 + 156 = **₹216** (½)

22. Without actual division, decide whether $\frac{27}{200}$ has a terminating decimal expansion. If it does, find the number of decimal places.

$200 = 2^3 \times 5^2 \Rightarrow$ terminating (1). Decimal places = greater of the powers 3 and 2 = **3** (1).

O R

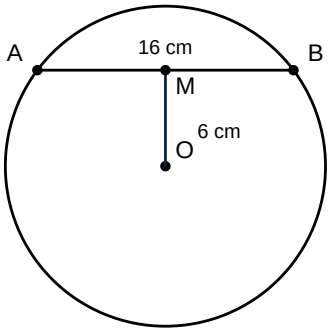
Find two rational numbers between $\frac{3}{7}$ and $\frac{4}{7}$.

3/7 = 24/56 and 4/7 = 32/56 (1). Two rational numbers: **25/56 and 26/56** (1). Any two valid numbers.

23. Factorise: $4x^2 - 20xy + 25y^2$.

$= (2x)^2 - 2(2x)(5y) + (5y)^2$ (1) = **$(2x - 5y)^2$** (1)

24. In the figure, O is the centre of the circle, chord AB = 16 cm and OM ⊥ AB with OM = 6 cm. Find the radius of the circle.



OM ⊥ AB bisects the chord: AM = 8 cm (1)
$OA^2 = OM^2 + AM^2 = 36 + 64 = 100 \Rightarrow$ **radius = 10 cm** (1)

25. Find the second, third and fourth terms of the sequence defined by $t_1 = 1$ and $t_n = t_{n-1} + 2n$ for $n \geq 2$.

$t_2 = 1 + 4 = 5$, $t_3 = 5 + 6 = 11$, $t_4 = 11 + 8 = 19$. (1 method, 1 values)

SECTION C — Short Answer (3 marks each) — 6 questions · 18 marks

Internal choice is provided in two questions.

26. Show that $2.142857142857\ldots = \frac{15}{7}$.

Let $x = 2.142857142857\ldots$; six digits repeat $\Rightarrow 10^6x = 2142857.142857\ldots$ (1)
$10^6x - x = 2142855 \Rightarrow 999999x = 2142855$ (1)
$x = 2142855/999999 =$ **15/7** (dividing by 142857) (1)

27. A shopkeeper sold 120 shirts:

Size	S	M	L	XL
Shirts sold	18	42	36	24

One shirt is chosen at random from those sold. Find the probability that it is of size (i) M, (ii) L or XL, (iii) not S.

> (i) 42/120 = **7/20** (1) (ii) 60/120 = **1/2** (1) (iii) 102/120 = **17/20** (1)

OR

The number of siblings of 50 students was recorded:

Siblings	0	1	2	3
Students	8	22	15	5

A student is chosen at random. Find the probability that the student has (i) exactly one sibling, (ii) at least two siblings, (iii) no sibling.

> (i) 22/50 = **11/25** (1) (ii) (15 + 5)/50 = **2/5** (1) (iii) 8/50 = **4/25** (1)

28. Find the coordinates of the point on the x-axis which is equidistant from A(2, −5) and B(−2, 9).

> Let the point be P(x, 0). $PA^2 = (x-2)^2 + 25$, $PB^2 = (x+2)^2 + 81$ (1)
> PA = PB ⇒ $x^2 - 4x + 29 = x^2 + 4x + 85$ (1) ⇒ $8x = -56$ ⇒ $x = -7$. Point = **(−7, 0)** (1)

29. Factorise: $x^3 + 6x^2 + 12x + 8$.

> $x^3 + 3\cdot x^2\cdot 2 + 3\cdot x\cdot 2^2 + 2^3$ (1) $= (x+2)^3$ (identity $(a+b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$) (1) = **(x + 2)(x + 2)(x + 2)** (1)

30. AB and CD are equal chords of a circle with centre O. OM ⊥ AB and ON ⊥ CD. Prove that OM = ON.

> The perpendicular from the centre bisects a chord ⇒ AM = ½AB, CN = ½CD ⇒ AM = CN (1)
> In right triangles OMA and ONC: OA = OC (radii), AM = CN ⇒ △OMA ≅ △ONC (RHS) (1) ⇒ **OM = ON** (CPCT) (1)

31. A triangular plot has sides 11 m, 60 m and 61 m. Find its area using Heron's formula.

> s = 66 (1); area = √(66 × 55 × 6 × 5) (1) = √108900 = **330 m²** (1)

OR

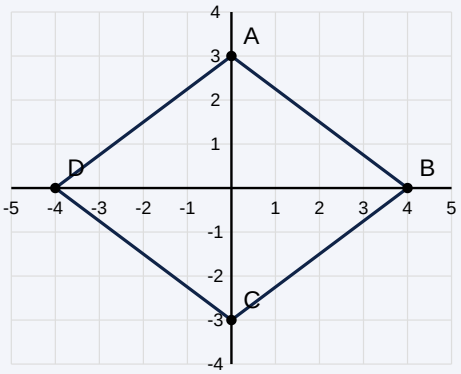
The sides of a triangle are in the ratio 5 : 12 : 13 and its perimeter is 60 cm. Find its area.

> Sides 10, 24, 26 cm (1); s = 30; area = √(30 × 20 × 6 × 4) (1) = √14400 = **120 cm²** (1)

SECTION D — Long Answer (5 marks each) — 4 questions · 20 marks

Internal choice is provided in two questions.

32. The corners of a rhombus-shaped garden are A(0, 3), B(4, 0), C(0, −3) and D(−4, 0) on a grid.
(a) Plot the points and join them in order. (2 marks)
(b) Find the lengths AB, BC, CD and DA. (1½ marks)
(c) Find the lengths of the diagonals and justify that ABCD is a rhombus but not a square. (1½ marks)



(a) Plotting (1½) + joining (½): (b) AB = √(16 + 9) = 5; BC = √(16 + 9) = 5; CD = 5; DA = 5 (1½)
(c) AC = 6, BD = 8 (½). All sides equal ⇒ rhombus; diagonals unequal ⇒ not a square (1)

33. Find how many natural numbers between 100 and 300 are multiples of 7, and find their sum. (2½ + 2½)

Numbers: 105, 112, …, 294 — AP with a = 105, d = 7, l = 294 (1)
n = (294 − 105)/7 + 1 = **28** (1½)
Sum = 28/2 × (105 + 294) = 14 × 399 = **5586** (2½)

34. In a survey of 400 families, the number of vehicles owned was recorded:

Vehicles	0	1	2	3 or more
Families	60	180	120	40

A family is chosen at random. Find the probability that it owns (i) no vehicle (1), (ii) exactly 2 vehicles (2), (iii) at least one vehicle (2).

(i) 60/400 = **3/20** (1) (ii) 120/400 = **3/10** (2) (iii) 340/400 = **17/20** (2)

O R

A box contains 5 green, 7 yellow and 8 purple candies. One candy is drawn at random. Find the probability that it is (i) green (1), (ii) yellow or purple (2), (iii) neither green nor yellow (2).

Total = 20. (i) 5/20 = **1/4** (1) (ii) 15/20 = **3/4** (2) (iii) 8/20 = **2/5** (2)

35. The length of a rectangular garden is 3 m more than twice its breadth, and its area is 65 m². Form an equation in the breadth x, solve it by factorisation and find the perimeter of the garden.

Breadth = x, length = 2x + 3 ⇒ x(2x + 3) = 65 ⇒ 2x² + 3x − 65 = 0 (1½)
2x² − 10x + 13x − 65 = (x − 5)(2x + 13) = 0 (1½) ⇒ x = 5 (negative rejected) (1)
Length = 13 m; perimeter = 2(5 + 13) = **36 m** (1)

O R

A student arranges 2 square tiles of side x units, 7 rectangular strips (x units by 1 unit) and 6 small square tiles (1 unit by 1 unit) to form one rectangle. Find its length and breadth in terms of x and its perimeter when x = 3.

Area = 2x² + 7x + 6 (1)
= 2x² + 4x + 3x + 6 = 2x(x + 2) + 3(x + 2) = **(2x + 3)(x + 2)** (2)
Length = 2x + 3, breadth = x + 2 (1). For x = 3: 9 and 5 ⇒ perimeter = 2(9 + 5) = **28 units** (1)

SECTION E — Case-Study Based Questions (4 marks each) — 3 questions · 12 marks

Read each case carefully and answer the questions that follow. Internal choice is provided in every 2-mark sub-part.

36. A gym charges a one-time joining fee of ₹500 and ₹300 per month. The total amount paid (in ₹) after m months is C = 500 + 300m.

(i) What is the total amount paid after 4 months?

> C = 500 + 1200 = **₹1700**

(ii) State the degree of C = 500 + 300m and its constant term.

> Degree = **1** (linear polynomial); constant term = **500** (joining fee).

(iii) After how many months will the total amount paid be ₹3500?

> 500 + 300m = 3500 (1) ⇒ m = **10 months** (1)

O R

(iii) Find the total amount after 7 months. Does the situation show linear growth or linear decay? Why?

> C = 500 + 2100 = **₹2600** (1). The amount increases at a constant rate (coefficient 300 > 0) ⇒ linear growth (1)

37. The area of a rectangular tabletop in a school workshop is given by the polynomial $x^2 + 9x + 20$ square units, where x is a positive number.

(i) Factorise $x^2 + 9x + 20$.

> $x^2 + 4x + 5x + 20 = (x + 4)(x + 5)$

(ii) If the breadth of the tabletop is (x + 4) units, write its length.

> Length = **(x + 5) units**

(iii) Find the perimeter of the tabletop when x = 3.

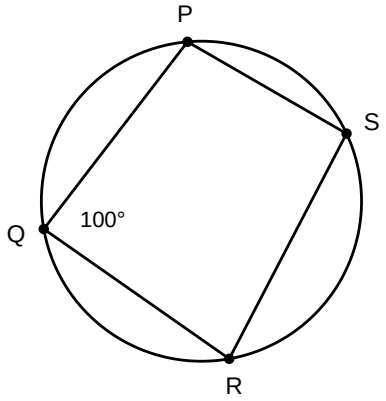
> Length = 8, breadth = 7 (1) ⇒ perimeter = 2(8 + 7) = **30 units** (1)

O R

(iii) Using the polynomial, find the area when x = 10. Verify with (x + 4)(x + 5).

> 100 + 90 + 20 = **210** (1); 14 × 15 = 210 ✓ (1)

38. A circular running track has four floodlight poles P, Q, R and S on its boundary, forming a cyclic quadrilateral PQRS. ∠PQR = 100° and ∠QPS = 85°. The diameter of the track is 50 m.



(i) Find ∠PSR.

> Opposite angles of a cyclic quadrilateral are supplementary ⇒ ∠PSR = 180° − 100° = **80°**

(ii) Find ∠QRS.

> ∠QRS = 180° − ∠QPS = 180° − 85° = **95°**

(iii) A straight path 40 m long is a chord of the track. Find its distance from the centre.

> r = 25 m; half chord = 20 (½). d = √(25² − 20²) (1) = **15 m** (½)

O R

(iii) Another straight path is a chord at a distance of 20 m from the centre. Find its length.

> Half chord = $\sqrt{(25^2 - 20^2)} = 15$ (1) ⇒ length = **30 m** (1)